Homework 7

October 28, 2026

Problem 1

Let $\mathcal{V}$ be a linear space over F, $\mathscr{A} \in L(\mathcal{V})$, $W \subset \mathcal{V}$ be $\mathscr{A}$-invariant, and $\psi(\lambda)$ be the minimal polynomial of $\mathscr{A}$ at W. Show that there exists $x \in W$ such that the minimal polynomial of $\mathscr{A}$ at x is $\psi(\lambda)$.

Problem 2

Let $\mathscr{A} \in L(C_n[\lambda])$ be defined as

$$\mathscr{A}[f(\lambda)] = f(\lambda + a), \qquad \forall\, f(\lambda) \in C_n[\lambda],$$

for a nonzero constant $a \in C$; its matrix representation A with respect to the basis $1, \lambda, \lambda^2, \ldots, \lambda^n$ was found in Homework 5. Find out

(1) the Smith canonical form S of the eigenmatrix $\lambda E - A$, and the Jordan canonical form J of A;

(2) the transition matrix X that reduces A to the Jordan canonical form J, i.e. $X^{-1}AX = J$.

Problem 3

Let E be an identity matrix with appropriate dimension,

$$\lambda + 3, \quad \lambda + 3, \quad (\lambda + 3)^2, \quad \lambda + 2 + 5j, \quad \lambda + 2 - 5j, \quad (\lambda + 2 + 5j)^2, \quad (\lambda + 2 - 5j)^2$$

are elementary divisors for $\lambda E - A$.

(1) Please find out the Smith canonical form S for $\lambda E - A$, the natural normal form N and the Jordan canonical form J for A.

(2) Let $\lambda_1 = -3$, $\lambda_2 = -2 + 5j$, $\lambda_3 = -2 - 5j$ and

$$T(\lambda_i) \triangleq \big\{x : \text{ there exists } k \in \mathcal{Z}_+ \text{ such that } (A - \lambda_i E)^k x = 0\big\},$$

$\mathcal{Z}_{+,0}$ is the set of non-negative integers, show that $T(\lambda_i) = \ker[(A - \lambda_i E)^2]$, $i = 1, 2, 3$, and determine the corresponding dimensions $\dim[T(\lambda_i)]$, $i = 1, 2, 3$.

Problem 4

Given a matrix A whose elementary divisor are

$$(\lambda+1), \quad (\lambda+1)^2, \quad (\lambda+1+j), \quad (\lambda+1-j), \quad (\lambda+1)^2, \quad (\lambda+1+j)^2, \quad (\lambda+1-j)^2.$$

(1) Find out the Smith canonical form S for $\lambda E - A$;

(2) Find out the Natural normal form N, the complex Jordan canonical form J_{complex} and the real Jordan canonical form J_{real} for A;

(3) The eigenvalues of matrix A are denoted as $\lambda_1 = -1$, $\lambda_2 = -1-j$, $\lambda_3 = -1+j$. For λ_i, $i = 1, 2, 3$, give the geometric multiplicity q_i and algebraic multiplicity $\hat{n}_i$;

(4) Find out $\dim[T(\lambda_i)]$, $i = 1, 2, 3$, where

$$T(\lambda_i) \triangleq \big\{x : \text{ there exists } k \in \mathcal{Z}_+ \text{ such that } (A - \lambda_i E_n)^k x = \mathbf{0}\big\}.$$